Landscape systems: reading notes

Working summaries for discussion - not a reading-completion log or a completed archaeological interpretation.

The immediate change is to consider terrain, water, productive land and dispersed settlement together. The original comparison boxes remain useful for consistent satellite statistics, but they are not archaeological landscape boundaries.

Van Liere 1980: start with landform

Traditional Water Management in the Lower Mekong Basin, pp. 265-280, especially Figs. 2 and 4-7. The useful distinction is between landform settings and ways of managing water: floodplain use, bunded-field retention and receding-flood cultivation should not be treated as one agricultural class. A green field near water is not, by itself, evidence of irrigation.

Method implication: examine the landscape position and configuration of fields and water features before proposing their function. Van Liere's strong separation of monumental/religious and agricultural hydraulic works is an argument to compare with other evidence, not a rule for our classifier.

Van Liere 1989: a Sukhothai-specific lead

Mon-Khmer Approaches to the Environment, particularly pp. 152-154 and Figs. 9.6-9.7. The Sukhothai discussion distinguishes a low terrace/fossil alluvial fan and hill-fed water supply from the Yom River lowlands. This gives a more specific starting point than applying a general Lower Mekong model to the whole study area.

Method implication: inspect hills, fan/terrace surfaces and river lowlands separately. Historical or ethnic interpretations in the chapter need corroboration; geometry alone cannot establish them. The Isanapura example is Sambor Prei Kuk, not Srei Santhor.

Evans et al. 2007: move beyond the monument

A comprehensive archaeological map ... at Angkor, pp. 14277-14282, especially the Mapping section and Fig. 2. The authors define their 2,848 km2 survey using watershed boundaries and combine multiple image, map and survey sources. Their mapped settlement includes dispersed temples, ponds, occupation areas and infrastructure beyond the major monuments.

Method implication: preserve our approximately 2,000 km2 comparison core, while defining a separate catchment/settlement context. Their archaeological confidence cannot simply be transferred to our coarser imagery. Their term 'linear feature' is also useful where a canal-versus-road function is uncertain.

Hydraulic interpretations to compare

The papers do not all make the same argument. Their differences guide questions, rather than supply labels for a new map.

Groslier 1979: read levels and connections

La cite hydraulique angkorienne: exploitation ou surexploitation du sol?, maps 1-7 on pp. 196-202. Rivers, spot heights, reservoirs and dispersed temples are read together. Maps 2-6 distinguish successive historical configurations rather than showing every feature as contemporary.

Method implication: examine plausible source-to-storage-to-downstream relationships. The mapped irrigation areas are historical interpretations, not observed 2025 agricultural polygons. Map 7's reconstructed archaeological lake margin near +16 m must not become a modern DSM threshold without checking datum, historical lake conditions and mapping uncertainty.

The debate is part of the method

Van Liere 1980 challenges Groslier's hydraulic interpretation. Evans et al. 2007 report later mapping and survey evidence for an integrated network, without resolving every feature's function or date. Two seasonal satellite composites do not settle that debate, and proximity to an old reservoir does not demonstrate continued productive benefit.

Hawken and Cobo Castillo: productive open space

Angkor's Agrarian Economy: A Socio-Ecological Mosaic, opening pp. 338-340. The chapter treats productive open spaces as part of the dispersed settlement rather than empty space between monuments. It emphasises varied strategies and landscape patches rather than one uniform agrarian system.

Method implication: keep field configuration and landform central. Similar-looking fields need not share the same date, management or hydrological role. Only the opening section has been reviewed for these notes.

Evans lidar manuscript: document interpretation

Selected sections, particularly PDF pp. 17-18, 26 and 36-37, of The Predicament of Archaeological Lidar. Page 26 stresses documentation such as interpreter identity and digitising scale. Appropriate data, existing knowledge and explicit interpretation records matter.

Method implication: attach source, scale, interpretation, confidence and supporting evidence to future features. The present Copernicus surface model is not archaeological lidar. The manuscript's publication year is not verified from the supplied copy; the filename alone is not sufficient.

Terrain audit and next steps

Measured from the existing cached surface models. These statistics describe our analytical boxes, not whole archaeological landscapes.

Region	P05 (m)	Median (m)	P95 (m)	P95-P05 (m)
Angkor	5.3	17.0	55.6	50.3
Srei Santhor	3.6	8.3	13.2	9.6
Sukhothai	45.1	60.0	380.3	335.2
Prambanan	52.1	191.4	748.2	696.1

P05 and P95 are the fifth and ninety-fifth elevation percentiles. Their difference describes broad variation; it is not a measurement of terrain accuracy.

Prambanan has the largest broad variation in these boxes. The Sukhothai box also includes substantial high ground, so it should not be treated as a uniformly flat plain. Srei Santhor has a negative elevation tail (minimum about -17 m) requiring investigation against water coverage, editing masks and another terrain source; negative values alone do not prove an error.

Limits that matter before drainage modelling

The source is Copernicus GLO-30, nominally about 30 m, resampled to the 20 m analysis grid. It is a surface model, including vegetation/building effects, not a bare-earth lidar terrain model. Product documentation specifies EGM2008 heights in metres, while local files encode only horizontal UTM CRSs. No vertical transformation or site-specific elevation validation has been performed.

The existing mosaics have finite values throughout the boxes, but this does not prove accurate terrain. The old preparation used zero as a fill/validity distinction; native DEMs and quality masks need to be retained and checked before routing or fusion. Existing cached tile files are already warped.

Proposed sequence

1. Acquire a wider native Angkor DEM context covering headwaters and outlets, rather than another temple-centred box.
2. Compare an additional suitable elevation source, including datum, masks, forest/urban bias and tile seams; do not average products by default.
3. Establish checked catchments and broad profiles, retaining unconditioned and conditioned surfaces separately.
4. Add a sourced feature register, then distinguish proximity, plausible terrain connection and documented hydraulic connection.
5. Apply site-specific reasoning to the other three regions, keeping the original comparison areas separate.

The source specifications are documented at Copernicus Data Space. Published product accuracy is not a local error measurement.

Coverage and references

A bounded literature-to-method summary. Remaining reading and analytical gaps are retained explicitly.

Coverage of the current notes

Van Liere 1980: text and relevant landform/Angkor maps reviewed. Van Liere 1989: text review with relevant mapped examples and the Sukhothai page inspected visually; some extraction is noisy. Evans et al. 2007: article text and regional map reviewed. Groslier 1979: all seven map plates and captions reviewed visually, but the full scanned French text has not yet been fully reviewed. The Hawken chapter and longer Evans manuscript are selected-section reviews, not full-paper completion claims.

These are working summaries for discussion, not a personal reading-completion log. No new catchment delineation, DEM fusion, validated drainage model, sourced temple inventory, feature dating or accepted 2025 agricultural-area estimate is claimed.

References supplied for the project

Van Liere, W. J. (1980). Traditional Water Management in the Lower Mekong Basin. World Archaeology 11(3), 265-280.

Van Liere, W. J. (1989). Mon-Khmer Approaches to the Environment. Chapter 9, Culture and Environment in Thailand, Siam Society symposium volume. Relevant examples: pp. 152-154.

Evans, D., et al. (2007). A comprehensive archaeological map of the world's largest preindustrial settlement complex at Angkor, Cambodia. PNAS 104(36), 14277-14282. doi:10.1073/pnas.0702525104.

Groslier, B.-P. (1979). La cite hydraulique angkorienne: exploitation ou surexploitation du sol? Bulletin de l'Ecole francaise d'Extreme-Orient 66, 161-202. Maps 1-7: pp. 196-202.

Hawken, S., and Cobo Castillo, C. Angkor's Agrarian Economy: A Socio-Ecological Mosaic. Supplied chapter, opening pp. 338-340. doi:10.4324/9781351128940-24.

Evans, D. The Predicament of Archaeological Lidar: Perspectives from Southeast Asia. Supplied manuscript; publication details to be confirmed.

References are to printed article/chapter pages unless explicitly identified as PDF pages. The recipient already holds the source papers; they are not duplicated in this upload package.